A Proof

This section proves the propositions and corollary used to derive the exact and approximate seasonal-component extraction methods. The following subsections establish phase-wise independence, characterize optimal candidates, relate the relaxed and original objectives, and justify the approximation strategy.

A.1 Proposition 3.1

Proof. By substituting r_i and τ_i into the STCD optimization objective, we obtain:

$$\begin{aligned}
&\sum_{i=1}^{m}\mathcal{B}\left(\mathcal{Z}\left(s_i\right)\right)+\sum_{i=2}^{n}\mathcal{B}\left(\mathcal{Z}\left(r_i\right)\right)+\epsilon\\
=&\sum_{i=1}^{m}\mathcal{B}\left(\mathcal{Z}\left(s_i\right)\right)+\sum_{i=2}^{n}\mathcal{B}\left(\mathcal{Z}\left(\Delta y'_i-s_{i \bmod m}\right)\right)+\epsilon\\
=&\sum_{i=1}^{m}\mathcal{B}\left(\mathcal{Z}\left(s_i\right)\right)+\sum_{i=1}^{m}\sum_{j=0}^{\lfloor\frac{n-i}{m}\rfloor}\mathcal{B}\left(\mathcal{Z}\left(\Delta y'_{i+j\times m}-s_i\right)\right)+\epsilon\\
=&\sum_{i=1}^{m}\left(\mathcal{B}\left(\mathcal{Z}\left(s_i\right)\right)+\sum_{j=0}^{\lfloor\frac{n-i}{m}\rfloor}\mathcal{B}\left(\mathcal{Z}\left(\Delta y'_{i+j\times m}-s_i\right)\right)\right)+\epsilon.
\end{aligned}$$

It can be observed that the optimization objective of STCD is the sum of the optimization objectives of all STCD-i subproblems. Therefore, the seasonal component s_i can be independently solved through each STCD-i subproblem, with the optimal solution being the sum of the optimal values from each subproblem. □

A.2 Proposition 3.2

Proof. Since $s_i \in \mathbb{Z}$, $\|s_i - s_i^*\| \le \delta$, $\bar{f}(s_i^*) \ge \bar{f}(s_i)$ equals to $\bar{f}(s_i+1) \ge \bar{f}(s_i)$, $\bar{f}(s_i-1) \ge \bar{f}(s_i)$. Suppose there exists a $s_i' \notin C_i$ and $\bar{f}(s_i+1) \ge \bar{f}(s_i)$, $\bar{f}(s_i'-1) \ge \bar{f}(s_i')$, we discuss the value range of s_i' as follows. For simplicity, we denote s' as s in the following discussion.

(1) If $s_i > \sup(C_i)$, then $\Delta y_{i+j\times m} - s_i > 0$ and $s_i > 0$,

$$\bar{f}(s_i) = \log_2(s_i) + \sum_{j=0}^{\lfloor\frac{n-i}{m}\rfloor}\log_2\left(\Delta y'_{i+j\times m}-s_i\right) + 2\times\left(\lfloor\frac{n-i}{m}\rfloor+2\right).$$

Taking the derivative of this function yields:

$$\bar{f}'(s_i) = \frac{\ln(2)}{s_i} + \sum_{j=0}^{\lfloor\frac{n-i}{m}\rfloor}\frac{\ln(2)}{\Delta y'_{i+j\times m}-s_i}.$$

For $s_i > \sup(C_i)$, it always holds that $\bar{f}'(s_i) > 0$, and thus $\bar{f}(s_i)$ is monotonically decreasing. That is, for any s_i, $\bar{f}(s_i+1) < \bar{f}(s_i)$, $\bar{f}(s_i-1) > \bar{f}(s_i)$. This is a contradiction, since there should exist a s_i where $\bar{f}(s_i+1) \ge \bar{f}(s_i)$, $\bar{f}(s_i-1) \ge \bar{f}(s_i)$.

(2) If $s_i < \inf(C_i)$, then $\Delta y_{i+j\times m} - s_i < 0$ and $s_i < 0$,

$$\begin{aligned}
\bar{f}(s_i) =&\log_2(-2s_i-1) + \sum_{j=0}^{\lfloor\frac{n-i}{m}\rfloor}\log_2\left(-2\times\left(\Delta y'_{i+j\times m}-s_i\right)-1\right)\\
&+\lfloor\frac{n-i}{m}\rfloor+2.
\end{aligned}$$

Taking the derivative of this function yields:

$$\bar{f}'(s_i) = \frac{2\times\ln(2)}{2s_i+1} + \sum_{j=0}^{\lfloor\frac{n-i}{m}\rfloor}\frac{2\times\ln(2)}{2\times(\Delta y'_{i+j\times m}-s_i)+1}.$$

For $s_i < \inf(C_i)$, it always holds that $\bar{f}'(s_i) > 0$, and thus $\bar{f}(s_i)$ is monotonically increasing. That is, for any s_i, $\bar{f}(s_i+1) > \bar{f}(s_i)$, $\bar{f}(s_i-1) < \bar{f}(s_i)$. This is a contradiction, since there should exist a $s_i \notin \{\Delta y_{i+j\times m} \mid j = 0, 1, \cdots, \lfloor\frac{n-i}{m}\rfloor\} \cup \{0\}$ and $\bar{f}(s_i+1) \ge \bar{f}(s_i)$, $\bar{f}(s_i-1) \ge \bar{f}(s_i)$.

(3) If $\inf(C_i) < s_i < \sup(C_i)$, without loss of generality, we assume $0, \Delta y_{i+0\times m}, \cdots, \Delta y_{i+k\times m} < s_i < \Delta y_{i+(k+1)\times m}, \cdots, \Delta y_{i+\lfloor\frac{n-i}{m}\rfloor\times m}$, then

$$\begin{aligned}
\bar{f}(s_i) =&\log_2(s_i) + \log_2(s_i-\Delta y_{i+0\times m}) + \cdots + \log_2(s_i-\Delta y_{i+k\times m})\\
&+\log_2(-2\times(s_i-\Delta y_{i+(k+1)\times m})-1) + \cdots\\
&+\log_2(-2\times(s_i-\Delta y_{i+\lfloor\frac{n-i}{m}\rfloor\times m})-1).
\end{aligned}$$

Furthermore, we can derive:

$$\begin{aligned}
\bar{f}(s_i+1)-\bar{f}(s_i) =& \log_2(1+\frac{1}{s_i}) + \log_2(1+\frac{1}{s_i-\Delta y_{i+0\times m}}) + \cdots\\
&+\log_2(1+\frac{1}{s_i-\Delta y_{i+k\times m}}) + \cdots\\
&+\log_2(1+\frac{1}{-2\times(s_i-\Delta y_{i+\lfloor\frac{n-i}{m}\rfloor\times m})-1})\\
\bar{f}(s_i+1)-\bar{f}(s_i) =& \log_2(1+\frac{1}{s_i})(1+\frac{1}{s_i-\Delta y_{i+0\times m}})\cdots\\
&(1+\frac{1}{s_i-\Delta y_{i+k\times m}})\cdots\\
&(1+\frac{1}{-2\times(s_i-\Delta y_{i+\lfloor\frac{n-i}{m}\rfloor\times m})-1}).
\end{aligned}$$

Similarly,

$$\begin{aligned}
\bar{f}(s_i-1)-\bar{f}(s_i) =& \log_2(1-\frac{1}{s_i}) + \log_2(1-\frac{1}{s_i-\Delta y_{i+0\times m}}) + \cdots\\
&+\log_2(1-\frac{1}{s_i-\Delta y_{i+k\times m}}) + \cdots\\
&+\log_2(1-\frac{1}{-2\times(s_i-\Delta y_{i+\lfloor\frac{n-i}{m}\rfloor\times m})-1})\\
\bar{f}(s_i-1)-\bar{f}(s_i) =& \log_2(1-\frac{1}{s_i})(1-\frac{1}{s_i-\Delta y_{i+0\times m}})\cdots\\
&(1-\frac{1}{s_i-\Delta y_{i+k\times m}})\cdots\\
&(1-\frac{1}{-2\times(s_i-\Delta y_{i+\lfloor\frac{n-i}{m}\rfloor\times m})-1}).
\end{aligned}$$

Considering $\bar{f}(s_i+1)-\bar{f}(s_i) \ge 0$ and $\bar{f}(s_i-1)-\bar{f}(s_i) \ge 0$, then

$$\begin{aligned}
&(1+\frac{1}{s_i})(1+\frac{1}{s_i-\Delta y_{i+0\times m}})\cdots(1+\frac{1}{s_i-\Delta y_{i+k\times m}})\cdots\\
&(1+\frac{1}{-2\times(s_i-\Delta y_{i+\lfloor\frac{n-i}{m}\rfloor\times m})-1}) \ge 1,\\
&(1-\frac{1}{s_i})(1-\frac{1}{s_i-\Delta y_{i+0\times m}})\cdots(1-\frac{1}{s_i-\Delta y_{i+k\times m}})\cdots\\
&(1-\frac{1}{-2\times(s_i-\Delta y_{i+\lfloor\frac{n-i}{m}\rfloor\times m})-1}) \ge 1.
\end{aligned}$$

Multiplying the left-hand sides of these two equations yields:

$$(1-(\frac{1}{s_i})^2)(1-(\frac{1}{s_i-\Delta y_{i+0\times m}})^2)\cdots(1-(\frac{1}{s_i-\Delta y_{i+k\times m}})^2)\cdots$$
$$(1-(\frac{1}{-2\times(s_i-\Delta y_{i+\lfloor\frac{n-i}{m}\rfloor\times m})-1})^2)\leq 1.$$

This is a contradiction.

In summary, for any $s_i\in\mathbb{Z}$ such that there exists a δ, where for all s_i^* satisfying $\|s_i-s_i^*\|\leq\delta$, $\bar{f}(s_i^*)\geq\bar{f}(s_i)$, it always holds that $s_i\in C_i$. □

A.3 Proposition 3.3

Proof. Suppose that $\bar{f}(s_i^*)\geq\bar{f}(s_i)$ for all $s_i^*\in\mathbb{Z}$ and there exists a $s_i'\in\mathbb{Z}$ satisfies $f(s_i')\leq f(s_i)$. Since $s_i'\in\mathbb{Z}$, $\bar{f}(s_i')\geq\bar{f}(s_i)$. Considering that $f(s_i)=\mathcal{B}(\mathcal{Z}(s_i))+\sum_{j=0}^{\lfloor\frac{n-i}{m}\rfloor}\mathcal{B}\left(\mathcal{Z}\left(\Delta y'_{i+j\times m}-s_i\right)\right)$ and $\bar{f}(s_i)=\bar{\mathcal{B}}(\mathcal{Z}(s_i))+\sum_{j=0}^{\lfloor\frac{n-i}{m}\rfloor}\bar{\mathcal{B}}\left(\mathcal{Z}\left(\Delta y'_{i+j\times m}-s_i\right)\right)$, the two functions differ only in $\bar{\mathcal{B}}$ and $\mathcal{B}$, where $\bar{\mathcal{B}}$ incorporates a downsampling function compared to $\mathcal{B}$. Therefore, $f(s_i^*)\leq\bar{f}(s_i^*)$ for all $s_i^*\in\mathbb{Z}$. Since the downsampling function does not affect the relative magnitude of the function values, for any $f(s_i')\leq f(s_i)$, $\bar{f}(s_i')\leq\bar{f}(s_i)$. This is a contradiction.

In summary, if $s_i\in\mathbb{Z}$ satisfies $\bar{f}(s_i^*)\geq\bar{f}(s_i)$ for all $s_i^*\in\mathbb{Z}$, it always holds that $f(s_i^*)\geq f(s_i)$ for all $s_i^*\in\mathbb{Z}$. □

A.4 Proposition 3.5

Proof. Since $\bar{f}(s_i)$ is a continuous and non-monotonic function, it necessarily admits an optimal point $\bar{s}_i$ such that for all $s_i^*\in\mathcal{Z}$, $\bar{f}(\bar{s}_i)>\bar{f}(s_i^*)$. Considering that a global optimal point is also a local optimal point, i.e., there exists a δ such that for all s_i^{**} satisfying $\|\bar{s}_i-s_i^{**}\|\leq\delta$, $\bar{f}(s_i^{**})\geq\bar{f}(\bar{s}_i)$. From Proposition 3.2, we derive that $\bar{s}_i\in C_i$. Since $\bar{f}(\bar{s}_i)>\bar{f}(s_i^*)$, from Proposition 3.3, we derive that $f(\bar{s}_i)>f(s_i^*)$. In conclusion, there exists an $s_i\in C_i$, i.e., $\bar{s}_i$, such that for all $s_i^*\in\mathcal{Z}$, $\bar{f}(\bar{s}_i)>\bar{f}(s_i^*)$. □

A.5 Corollary 3.7

Proof. We derive a simplified form of $\bar{f}(s_i)$.

$$\begin{aligned}\bar{f}(s_i)=&\bar{\mathcal{B}}(\mathcal{Z}(s_i))+\sum_{j=0}^{\lfloor\frac{n-i}{m}\rfloor}\bar{\mathcal{B}}\left(\mathcal{Z}\left(\Delta y'_{i+j\times m}-s_i\right)\right)\\=&\sum_{c_j\in C_i\wedge c_j<s_i}\log_2(2s_i-2c_j-1)+\\&\sum_{c_j\in C_i\wedge c_j>s_i}\log_2(2c_j-2s_i)+\lfloor\frac{n-i}{m}\rfloor\\=&\sum_{c_j\in C_i\wedge c_j<s_i}\log_2(s_i-c_j-\frac{1}{2})+\\&\sum_{c_j\in C_i\wedge c_j>s_i}\log_2(c_j-s_i)+2\lfloor\frac{n-i}{m}\rfloor\end{aligned}$$

Thus, we obtain $\bar{f}(s_i)=\sum_{c\in C_i}\log_2\mid s_i-c\mid+\epsilon$, where ϵ is a constant.

□

Table 8: Complete list of notations.

Symbol	Description
y	Original time series, i.e., $y=(y_1,\ldots,y_n)$
y_i	The value of the time series at timestamp i
y_i'	Scaled integer value of y_i
n	Length of the time series
m	Period length of the time series
ρ	Scaling factor for converting real values into integers
$\Delta y_i'$	De-trended value, i.e., $y_i'-y_{i-1}'$
τ_i	Trend component of y_i
s_i	Seasonal component at phase i
r_i	Residual component of y_i
C_i	Candidate set for seasonal selection at phase i
$f(s_i)$	Original storage-cost objective for phase i
$\bar{f}(s_i)$	Continuous relaxation of $f(s_i)$
$S_v(\cdot), S_d(\cdot)$	Value and density scores
$\mathcal{Z}(\cdot)$	ZigZag encoding function for signed integers
$\mathcal{B}(\cdot)$	Bit-width function for encoded integer values
z_i	ZigZag-encoded residual value
b_i	Bit-width of z_i
w_i	Bit-packing width of the i-th block
ξ	Compressed byte stream
ϵ	Constant metadata overhead in compressed storage

A.6 Proposition 3.10

Proof. If the elements $c\in C_i$ follow a Gaussian distribution, we leverage two key properties of the Gaussian distribution: (1) symmetry around the center, and (2) higher point density near the center. Owing to symmetry, the optimal seasonal component s is located at the center of the distribution. Under the value criterion, the central point minimizes the deviation due to symmetry. Under the density criterion, the central point also attains the minimum value because it lies in the region of highest local density. Consequently, both criteria consistently identify the center as the optimal candidate, and the proposed method correctly selects the optimal seasonal component s. □

B Notations

This section lists the complete symbols used in the formulation, compression procedure, and theoretical analysis in Table 8.

C Pseudocode

This section presents the complete STCD compression and decompression procedures. Algorithm 1 constructs and encodes the seasonal and residual components, while Algorithm 2 reverses these steps to recover the original series.

C.1 Compression

Algorithm 1 first scales and differences the input series, then selects one seasonal value for each phase. It derives residuals from the selected seasonal values and serializes the seasonal values, residuals, and metadata into the compressed byte stream.

Algorithm 1: STCD Compression

Input: original time series $y \in \mathbb{R}^n$, period length m, scale factor ρ, variant mode $\in$ {STCD-E, STCD-A}
Output: compressed byte stream ξ

Scale y to integer values y' by Formula 2;
Compute the de-trended series Δy by Formula 3;
for $i \leftarrow 1$ *to* m **do**
  Construct the candidate set C_i;
  if mode = *STCD-E* **then**
    $s_i \leftarrow \arg\min_{c \in C_i} f(c)$;
  else
    Compute $S_v(c)$ and $S_d(c)$ for each $c \in C_i$;
    $s_i \leftarrow \arg\min_{c \in C_i} (S_v(c) + S_d(c))$;
for $i \leftarrow 1$ *to* n **do**
  $r_i \leftarrow \Delta y_i - s_{i \bmod m}$;
Encode r and s by ZigZag using Formula 4;
Compute residual bit-widths b by Formula 5;
Serialize residual values by bitwise varint using Formula 6;
Serialize residual lengths by bit-packing using Formulas 7;
Serialize encoded seasonal components and metadata into ξ;
return ξ;

Algorithm 2: STCD Decompression

Input: compressed byte stream ξ
Output: original time series $y \in \mathbb{R}^n$

Read metadata from ξ, including y'_0, n, m, ρ, block bit-widths, and seasonal components s;
Decompress residual lengths, ZigZag-encoded residuals, and residuals by Formula 8;
for $i \leftarrow 1$ *to* n **do**
  Decompress Δy_i by Formula 9;
Decompress the scaled series y' and original series y by Formula 10;
return y;

C.2 Decompression

Algorithm 2 reads the stored metadata and seasonal values, decodes the residuals, and reconstructs the delta sequence. It then restores the scaled series by cumulative summation and inversely scales it to obtain the original series.

D Detailed Complexity Analysis

In this section, we analyze the time complexity of STCD for a time series of length n and period length m. The compression procedure consists of two parts: the standard linear encoding steps and the STCD-specific seasonal extraction step. The standard steps, including scaling, de-trending, ZigZag encoding, bitwise varint encoding, and bit-packing, each scan the data once and therefore require $O(n)$ time.

The dominant cost of STCD comes from extracting the seasonal components. Since the seasonal components are shared by values with the same phase, STCD solves m phase-wise subproblems. For STCD-E, the i-th subproblem contains approximately n/m same-phase values. The exact method evaluates each candidate in C_i against these same-phase values, resulting in $O((n/m)^2)$ time per phase and $O(n^2/m)$ time over all phases. Therefore, the total compression time of STCD-E is $O(n^2/m)$.

For STCD-A, the seasonal component is selected using the value and density scores. For each phase, candidates are sorted once, and the scores are computed from the sorted order. This costs $O((n/m)\log(n/m))$ time per phase and $O(n\log(n/m))$ time over all phases. Therefore, the total compression time of STCD-A is $O(n\log(n/m))$.

During decompression, STCD reads the metadata, decodes the residual lengths and residual values, adds the seasonal components back to the residual the residual components, and restores the trend components by cumulative summation. Each step performs a linear scan of the data. Therefore, the decompression time of STCD is $O(n)$.

E Experiments

This section provides supplementary experimental details and results. Section E.1 describes the datasets and system parameters, and the following Dataset Examples Section E.2 illustrates representative dataset characteristics. Section E.3.1 reports decompression-time comparisons, and Section E.3.2 evaluates sensitivity to different same-phase difference distributions. Sections E.4.1 and E.4.2 evaluate scalability and block-size sensitivity, respectively. Sections E.5.1 and E.5.2 present the classification and clustering results.

E.1 Experimental Setup

This subsection details the datasets and system parameters used in the experiments.

E.1.1 Datasets. Table 2 lists the time series datasets used in our experiments, including public data and data collected by our industrial partners. As for public data, Power [41] contains records of power consumption from three different distribution networks in Tetouan city, located in northern Morocco. Temp [52] contains records of air temperatures of a wind farm. Weather, Traffic, and Electricity datasets, sourced from Informer [61], have been widely used to benchmark long-sequence time series forecasting. Wind-dir comes from the NEON 2D wind speed and direction dataset [37], using wind direction angles in $[0°, 360°]$. Stocks-USA comes from the INFORE Monetary (Stocks) dataset [20].

We also employ some real-world datasets from our industrial partners. GHI, short for Global Tilted Irradiance, records the total solar radiation received on a horizontal surface at a specific location. It is collected by our meteorological research partners. Insurance contains records of insurance transactions, while the Telecom dataset includes records of telecom traffic. Both datasets are provided by our industry partners.

E.1.2 System Parameters. To ensure a fair comparison, all compressors use the same system parameters. The default block size is 1,024, and each default input page contains at most 1,025 values. All inputs use the `DOUBLE` type with a fixed sensor precision of two decimal places. Every encoder uses the same page boundaries. The

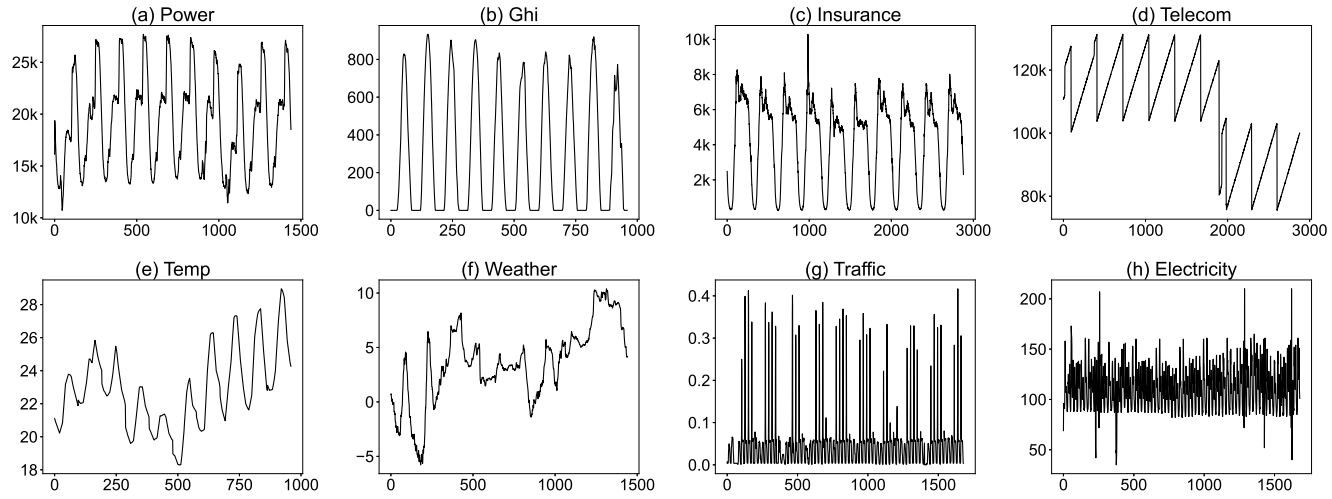


Figure 8: Example of datasets, where dataset Weather is a representative without clear seasonal features.

Table 9: Decompression time of compression methods (ns per value). Best results are highlighted in bold and second-best results are underlined.

Dataset	STCD-E	STCD-A	SPRINTZ	Gorilla	Chimp	BUFF	FLEA	Elf	ALP	Serf	LZ4	Snappy
Power	53.0	48.0	29.0	14.0	26.0	24.0	96.0	66.0	35.0	60.0	26.0	**10.0**
GHI	12.0	12.0	18.5	8.5	15.0	12.0	40.5	27.5	16.5	28.0	11.0	**6.5**
Insurance	14.0	15.0	23.0	13.0	19.0	15.0	27.0	40.0	19.0	58.0	16.0	**10.0**
Telecom	12.0	12.0	21.0	**8.0**	18.0	13.0	25.0	34.0	18.0	23.0	17.0	9.0
Air-Temp	13.0	12.0	22.0	14.0	17.0	12.0	30.0	36.0	14.0	55.0	14.0	**8.0**
Weather	9.0	9.0	19.0	**6.0**	12.0	8.0	31.0	15.0	8.0	12.0	8.0	6.0
Traffic	45.0	45.0	23.0	13.0	20.0	13.0	63.0	42.0	16.0	61.0	19.0	**8.0**
Electricity	12.0	12.0	23.0	**7.0**	18.0	9.0	34.0	31.0	11.0	22.0	13.0	7.0
Wind-dir	15.7	15.7	24.2	14.9	19.1	14.7	30.8	45.6	20.7	63.9	16.8	**10.1**
Stocks-USA	10.8	10.6	25.0	**7.2**	15.6	8.7	25.4	25.6	11.8	18.1	22.2	8.3
Household	15.2	15.0	18.4	9.3	14.4	10.9	63.1	26.4	12.1	31.4	29.8	**6.2**
PEMS-BAY	12.0	11.6	21.3	14.1	18.8	12.7	105.7	38.0	13.9	58.6	13.9	**6.6**
BDG2-Elec.	22.5	22.4	20.3	11.9	18.4	14.1	50.7	39.4	17.5	47.7	14.4	**7.2**
NREL-Solar	11.5	11.9	20.7	9.7	15.2	25.8	60.9	30.7	15.3	33.8	9.3	**5.8**
Median	12.5	12.0	21.6	10.8	18.0	12.8	37.2	35.0	15.6	40.8	15.2	**7.6**
Mean	18.4	18.0	22.0	10.8	17.6	13.8	48.8	35.5	16.3	40.9	16.5	**7.8**
Total	12.7	12.8	22.4	9.9	16.0	18.6	47.6	31.7	15.0	34.6	14.4	**7.1**

R2O2 R2D1 JVM uses an initial heap size of 256 MB and a maximum heap size of 1 GB.

E.2 Dataset Examples

To show the features of different datasets, such as trend and seasonality, we plot data points spanning 10 periods for each dataset in Figure 8. As shown, most datasets exhibit a certain degree of seasonality. The Weather dataset shows relatively weaker seasonality, the Power dataset exhibits significant nonlinear trends, and the Telecom dataset even presents abrupt trend changes.

E.3 Comparison with Existing Methods

This subsection provides additional comparisons with existing methods. Section E.3.1 reports decompression-time costs, and Section E.3.2 evaluates sensitivity to different same-phase difference distributions.

E.3.1 Decompression Time Cost. Table 9 reports the decompression time of all methods. STCD-E and STCD-A show similar decompression costs because they share the same decompression procedure. Their average decompression times are 18.4 ns and 18.0 ns per value, respectively. Although several lightweight methods such as Snappy

Table 10: Compression ratio of compression methods under different distributions of same-phase first differences. Best results are highlighted in bold and second-best results are underlined.

Dataset	STCD-E	STCD-A	SPRINTZ	Gorilla	Chimp	BUFF	FLEA	Elf	ALP	Serf	LZ4	Snappy
IoTDB-Benchmark Gaussian	**0.126**	0.126 (0.1%)	0.941 (86.6%)	1.018 (87.6%)	0.442 (71.4%)	1.018 (87.6%)	0.139 (9.4%)	0.489 (74.2%)	0.238 (47.1%)	0.925 (86.4%)	0.499 (74.7%)	0.499 (74.7%)
IoTDB-Benchmark Uniform	**0.123**	0.123 (0.5%)	0.941 (86.9%)	1.017 (87.9%)	0.442 (72.1%)	1.017 (87.9%)	0.139 (11.6%)	0.490 (74.8%)	0.238 (48.3%)	0.925 (86.7%)	0.500 (75.3%)	0.500 (75.3%)
IoTDB-Benchmark Student-t ($\nu = 3$)	**0.126**	0.126 (0.1%)	0.941 (86.6%)	1.018 (87.6%)	0.441 (71.4%)	1.018 (87.6%)	0.140 (10.0%)	0.489 (74.2%)	0.238 (47.1%)	0.925 (86.4%)	0.498 (74.7%)	0.498 (74.7%)
IoTDB-Benchmark Laplace	**0.126**	0.126 (0.1%)	0.942 (86.7%)	1.016 (87.6%)	0.441 (71.5%)	1.016 (87.6%)	0.139 (9.7%)	0.489 (74.3%)	0.238 (47.3%)	0.925 (86.4%)	0.497 (74.7%)	0.496 (74.7%)
IoTDB-Benchmark Skewed	**0.124**	0.125 (0.6%)	0.942 (86.8%)	1.017 (87.8%)	0.441 (71.8%)	1.017 (87.8%)	0.139 (10.3%)	0.489 (74.6%)	0.238 (47.8%)	0.925 (86.6%)	0.495 (74.9%)	0.494 (74.8%)
TSBS-IoT Gaussian	**0.168**	0.168 (0.1%)	0.959 (82.5%)	1.013 (83.4%)	0.490 (65.8%)	1.013 (83.4%)	0.196 (14.2%)	0.491 (65.8%)	0.269 (37.6%)	0.923 (81.8%)	0.582 (71.2%)	0.616 (72.8%)
TSBS-IoT Uniform	**0.172**	0.172 (0.1%)	0.959 (82.0%)	1.012 (83.0%)	0.490 (64.9%)	1.012 (83.0%)	0.196 (12.0%)	0.491 (65.0%)	0.269 (36.0%)	0.923 (81.4%)	0.583 (70.5%)	0.616 (72.1%)
TSBS-IoT Student-t ($\nu = 3$)	**0.160**	0.161 (0.2%)	0.959 (83.3%)	1.013 (84.2%)	0.490 (67.3%)	1.013 (84.2%)	0.196 (18.0%)	0.491 (67.3%)	0.269 (40.3%)	0.923 (82.6%)	0.582 (72.4%)	0.616 (73.9%)
TSBS-IoT Laplace	**0.163**	0.163 (0.2%)	0.959 (83.0%)	1.012 (83.9%)	0.490 (66.8%)	1.012 (83.9%)	0.196 (16.9%)	0.491 (66.9%)	0.269 (39.5%)	0.923 (82.4%)	0.583 (72.1%)	0.616 (73.6%)
TSBS-IoT Skewed	**0.160**	0.162 (0.9%)	0.959 (83.3%)	1.014 (84.2%)	0.490 (67.3%)	1.014 (84.2%)	0.196 (18.0%)	0.491 (67.3%)	0.269 (40.4%)	0.923 (82.6%)	0.582 (72.5%)	0.616 (74.0%)
Median	**0.143**	0.144 (0.2%)	0.950 (84.9%)	1.015 (85.9%)	0.466 (69.3%)	1.015 (85.9%)	0.168 (14.6%)	0.490 (70.8%)	0.254 (43.5%)	0.924 (84.5%)	0.541 (73.5%)	0.558 (74.3%)
Mean	**0.145**	0.145 (0.2%)	0.950 (84.7%)	1.015 (85.7%)	0.466 (68.9%)	1.015 (85.7%)	0.167 (13.5%)	0.490 (70.4%)	0.254 (42.9%)	0.924 (84.3%)	0.540 (73.2%)	0.557 (74.0%)
Total	**0.128**	0.128 (0.1%)	0.943 (86.4%)	1.017 (87.4%)	0.445 (71.3%)	1.017 (87.4%)	0.143 (10.8%)	0.489 (73.8%)	0.241 (46.8%)	0.925 (86.2%)	0.504 (74.6%)	0.506 (74.7%)

and Gorilla are faster in pure decompression time, the overall differences among most methods are small. This indicates that STCD achieves higher compression effectiveness without introducing a significant decompression overhead.

E.3.2 Sensitivity to Difference Distributions. For a potential user of STCD, the distribution of same-phase first differences is also relevant when deciding whether STCD-A is suitable. Proposition 3.10 proves that the value and density scores select the optimal seasonal value when the elements in C_i follow a Gaussian distribution. This condition is sufficient rather than necessary. Other distributions may also lead STCD-A to select the optimal value, but the proposition does not establish this result.

Changing the difference distribution changes the generated series and thus the compression ratios of all encoders. We retain the periodic templates of IoTDB-Benchmark and TSBS-IoT and generate random-walk remainders with Gaussian, uniform, Student-t with $\nu = 3$, Laplace, or skewed heavy-tailed innovations. These distributions represent symmetric light-tailed, bounded, symmetric heavy-tailed, sharply peaked heavy-tailed, and asymmetric heavy-tailed differences, respectively. For each phase, the generated same-phase first differences in C_i equal the periodic difference centre plus a quantized innovation from the selected distribution. We evaluate three fixed seeds for each distribution and apply every encoder to the same generated input.

Table 10 shows that STCD-E achieves the best compression ratio for all ten workload-distribution combinations, and STCD-A is consistently second. The compression-ratio difference between STCD-E and STCD-A is at most 0.9%. STCD-E reduces the ratio relative to FLEA by 9.4%–11.6% on IoTDB-Benchmark and by 12.0%–18.0% on TSBS-IoT. The Gaussian result is consistent with Proposition 3.10, while the uniform, Student-t, Laplace, and skewed heavy-tailed results show that STCD-A also remains effective outside this sufficient condition. Overall, the results verify that STCD-E achieves the best compression ratio under every tested difference distribution. R1.W2 R1.D2

E.4 Evaluation of Proposed Method

This subsection provides additional evaluations of the proposed method. Section E.4.1 evaluates scalability as data size increases, and Section E.4.2 evaluates the effect of database block size.

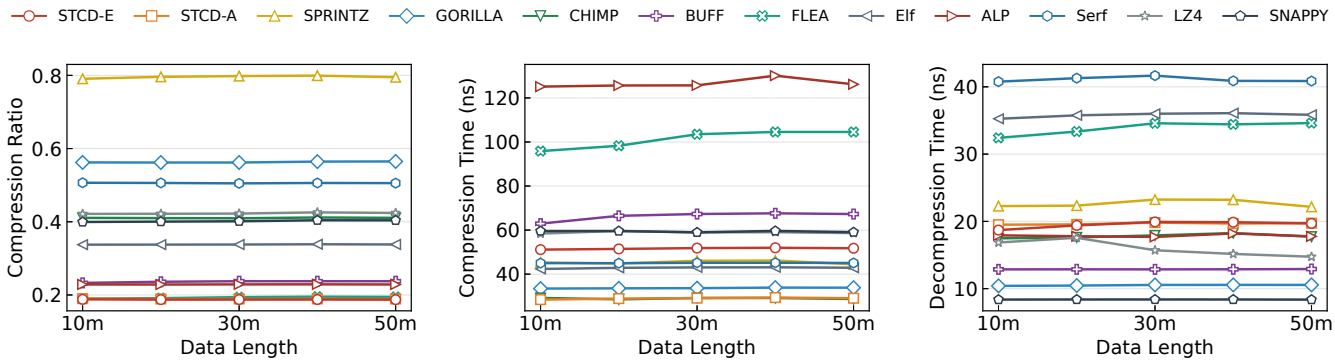


Figure 9: Varying data size. Each point reports the average over all 14 datasets.

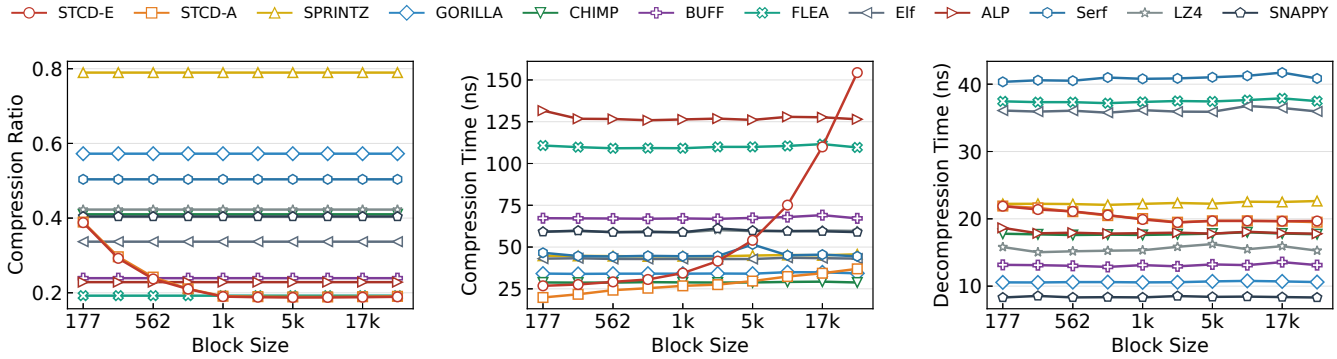


Figure 10: Varying block size. Each point reports the average over all ten datasets.

E.4.1 Scalability. Figure 9 evaluates scalability by reporting the average results over all datasets under different data sizes. Since compression and decompression time is measured as the processing time per value, it remains nearly unchanged as the data size increases. Overall, all compression methods, including STCD-E and STCD-A, show stable compression ratio and runtime across different data sizes. This indicates that STCD does not introduce additional per-value overhead on larger datasets.

R2.O2

R2.D1

E.4.2 Parameter Experiment. Overall, STCD is parameter-free and requires no user tuning. As mentioned in Section 4.1, databases typically store data in blocks. In this section, we analyze the impact of block size on performance.

Figure 10 evaluates the impact of database block size. Since STCD performs seasonal extraction at the block level, only STCD-E and STCD-A are directly affected by the block size. As the block size increases, both STCD-E and STCD-A achieve better compression ratios because larger blocks provide more data for seasonal-component estimation. However, larger blocks also increase compression time. Compared with STCD-E, the compression time of STCD-A grows much more slowly, showing that the approximation strategy effectively controls the block-size-related encoding overhead.

E.5 Comparison in Downstream Applications

This subsection reports additional downstream-application results. Section E.5.1 presents time-series classification, and Section E.5.2 presents time-series clustering.

E.5.1 Time Series Classification. For the classification task, each file is treated as an individual series labeled by its dataset name, with 80% of the data used for training and 20% for validation. The classification model jointly uses the stored seasonal and residual components as separate input representations. The seasonal component represents repeated periodic patterns, while the residual component represents deviations from these patterns. This separation allows the model to consider both regular periodic behavior and irregular variation instead of operating only on the original series. Figure 11(1) and (2) shows a trend similar to anomaly detection. STCD-E and STCD-A achieve the best efficiency by directly reading these compressed components without full decompression or re-decomposition. Their classification accuracy is comparable to most decomposition-based methods and is only clearly lower than RobustSTL, which incurs much higher time overhead. Overall, STCD provides a favorable balance between classification accuracy and efficiency.

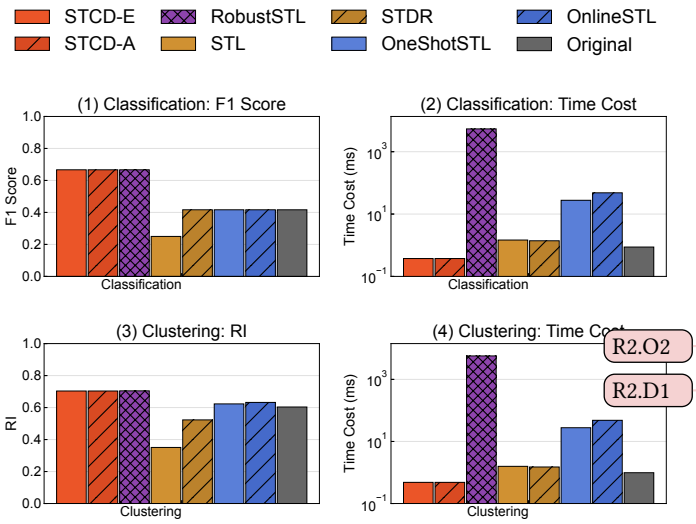


Figure 11: Performance of classification and clustering.

E.5.2 Time Series Clustering. Clustering follows the same dataset partitioning, label assignment, and component representation as classification. The clustering model jointly evaluates the seasonal component for similarity in repeated periodic patterns and the residual component for similarity in irregular variations. Figure 11(3) and (4) reports the clustering performance and time overhead. All methods achieve comparable clustering effectiveness, while STCD-E and STCD-A perform slightly better than most baselines. In terms of efficiency, STCD again achieves the lowest time overhead by directly reading the stored components and avoiding re-decomposition. These results demonstrate that the seasonal and residual components provide effective representations for the stated classification and clustering workflows.

R2.O2

R2.D1